UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

DEPARTMENT OF INFORMATION COMMUNICATION AND TECHNOLOGY

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Machine Learning and Data Mining 2

Labwork 1 Report

T1 – Principal component analysis

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I. Study the dataset

1. Dataset Overview

The datasets used in this project are the Breast Cancer Wisconsin (Diagnostic) (BC) dataset [1] and the Wine Recognition (WR) dataset [2]. The Breast Cancer dataset is computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. It describes characteristics of the cell nuclei present in the image and contains 31 dimensions, counting 30 features and 1 target variable. The Wine Recognition dataset involves the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. It contains 14 dimensions, 13 features, and 1 target variable.

2. Dataset Features

Features in BC and WR datasets are identified into different Type/Category/Data Type as shown in Table 1 and Table 2 respectively.Most features in the BC dataset, such as mean radius and mean texture, are classified as Continuous because they represent physical measurements that can take any real number value. Similarly, features in the WR dataset like alcohol and malic_acid are chemical concentrations, also classified as Continuous.In terms of Category, the features in both datasets are considered Quantitative because their values are measurable. The target variables (target in BC and target in WR) are Qualitative as they represent the categorization of data (Malignant/Benign for BC, and Cultivator 0/1/2 for WR).

Therefore, this classification, as shown in Table 1 and Table 2, allows us to better understand Type/Category/Date Type of these data in order to use them for machine learning.

Table 1: Breast Cancer (BC) dataset features identify

Feature	**Type**	**Category**	**Data Type**
Mean radius	Continuous	Quantitative	Numerical
Mean texture	Continuous	Quantitative	Numerical
Mean perimeter	Continuous	Quantitative	Numerical
Mean area	Continuous	Quantitative	Numerical
Mean smoothness	Continuous	Quantitative	Numerical
Radius error	Continuous	Quantitative	Numerical
Texture error	Continuous	Quantitative	Numerical
Perimeter error	Continuous	Quantitative	Numerical
Area error	Continuous	Quantitative	Numerical
Smoothness error	Continuous	Quantitative	Numerical
Compactness error	Continuous	Quantitative	Numerical
Target (Diagnosis)	Discrete	Qualitative	Categorical

Table 2: Wine Recognition (WR) dataset features identify

Feature	Type	Category	Data Type
Alcohol	Continuous	Quantitative	Numerical
Malic acid	Continuous	Quantitative	Numerical
Ash	Continuous	Quantitative	Numerical
Alcalinity of ash	Continuous	Quantitative	Numerical
Magnesium	Continuous	Quantitative	Numerical
Total phenols	Continuous	Quantitative	Numerical
Flavanoids	Continuous	Quantitative	Numerical
Nonflavanoid phenols	Continuous	Quantitative	Numerical
Proanthocyanins	Continuous	Quantitative	Numerical

3. Data Preparation

Data preparation is essential to ensure reliable analysis. A common issue is the difference in scales. In the BC dataset, mean area has values in the hundreds or thousands (mean ~655), while mean smoothness is a small fraction (mean ~0.096). This vast difference requires standardization (scaling) so that features with larger magnitudes do not dominate the Principal Component Analysis (PCA).

Similarly, the WR dataset contains proline (values >1000) and phenols (values <5), requiring the same scaling treatment.

4. Data Label

Firstly, we discuss the label in the datasets. In supervised learning, a label is the target variable. Both datasets that we are using in this project have clear labels. In the BC dataset, the target column is the label; it represents the diagnosis of the breast mass, categorized as either Malignant (0) or Benign (1). Moving on to the WR dataset, the label for this dataset is the target column. The values in this column, encoded as integers 0, 1, and 2, represent the specific cultivator that produced the wine, and it is used for classification tasks. In conclusion, labels are very important for supervised learning because they are the baseline for making comparisons with the prediction that the model returns.

5. Calculate Mean, Variance, Covariance and Correlation of the Datasets

In this section, we calculate the mean, variance, covarience, correlation of the datasets. To calculate the measurements, we used Python functions which have corresponse formula as below:

- **Mean:**
  - Python function: **.mean()**
  - Formula: $\bar{x} = \frac{1}{n}\Sigma x$
- **Variance:**
  - Python function: **.var()**
  - Formula: $var(x) = \sigma^2 = \frac{\Sigma(\bar{x}-x)^2}{n-1}$
- **Covariance:**
  - Python function: **.cov()**
  - Formula: $cov(x, y) = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{n-1}$
- **Correlation:**
  - Python function: **.corr()**

The table below shows the calculated mean, variance, covariance and correlation of each dataset respectively:

- **BC Dataset:**

Table 3: BC Dataset Mean and Variance

Feature	**Mean**	**Variance**
mean radius	14.127	12.419
mean texture	19.290	18.499
mean perimeter	91.969	590.440
mean area	654.889	123843.55
mean smoothness	0.096	0.0002
radius error	0.405172	0.076902
texture error	1.216853	0.304316
perimeter error	2.866059	4.087896
area error	40.337079	2069.431583
smoothness error	0.007041	0.000009
compactness error	0.025478	0.000321
quality	5.6360	0.6521

Table 4: BC dataset Covariance

	radius	texture	…	target	quality
radius	12.42	4.91	…	-1.25	0.1744

texture	4.91	18.50	…	-0.86	-0.0565
perimeter	85.45	34.44	…	-8.73	0.0356
area	1224.48	486.00	…	-120.74	0.0156
smoothness	0.01	-0.00	…	-0.00	-0.0049
compactness	0.09	0.05	…	-0.02	-0.4279
concavity	0.19	0.10	…	-0.03	-4.9172
concave pts	0.11	0.05	…	-0.01	-0.0002
symmetry	0.01	0.01	…	-0.00	-0.0072
fractal dim	-0.01	-0.00	…	0.00	0.0344
target	-1.25	-0.86	…	0.23	0.4098
quality	0.174424	-0.0565	…	0.4098	0.6522

Table 5: BC dataset Correlation

	radius	texture	…	target	quality
radius	1.000000	0.32	…	-0.73	0.12
texture	0.32	1.0000	…	-0.42	-0.39
perimeter	0.99	0.33	…	-0.74	0.22
area	0.99	0.32	…	-0.71	0.013
smoothness	0.17	-0.02	…	-0.36	-0.13

compactness	0.51	0.24	…	-0.60	-0.05
concavity	0.68	0.30	…	-0.70	-0.19
concave pts	0.82	0.29	…	-0.78	-0.17
symmetry	0.15	0.07	…	-0.33	-0.06
fractal dim	-0.31	-0.08	…	0.01	0.25
target	-0.73	-0.42	…	1.00	0.48
quality	0.12	-0.39	…	0.4762	1.00

- **WR Dataset:**

Table 6: WR Dataset Mean and Variance

Features	**Mean**	**Variance**
alcohol	13.0006	0.6591
malic_acid	2.3363	1.2480
ash	2.3665	0.0753
alcalinity_of_ash	19.4949	11.1527
magnesium	99.7416	203.9893
total_phenols	2.2951	0.3917
flavanoids	2.0293	0.9977
nonflavanoid_phenols	0.3619	0.0155
proanthocyanins	1.5909	0.3276

Table 7: WR Dataset Covariance

	alcohol	…	proanthocyanins
alcohol	0.66	…	0.06
malic_acid	0.09	…	-0.14
ash	0.05	…	0.00
alcalinity_of_ash	-0.84	…	-0.38
magnesium	3.14	…	1.93
total_phenols	0.15	…	0.22
flavanoids	0.19	…	0.37
nonflavanoid_phenols	-0.02	…	-0.03
proanthocyanins	0.06	…	0.33

Table 8: WR Dataset Correlation

	alcohol	…	proanthocyanins
alcohol	1.0000	…	0.14
malic_acid	0.09	…	-0.22
ash	0.21	…	0.01
alcalinity_of_ash	-0.31	…	-0.20
magnesium	0.27	…	0.24
total_phenols	0.29	…	0.61
flavanoids	0.24	…	0.65

nonflavanoid_phenols	-0.16	…	-0.37
proanthocyanins	0.14	…	1.00

6. Correlated Couple of Features each Dataset

With the results from table 5 and table 8 for the RWQ and CCS datasets respectively, In the BC dataset, the highest correlation between Mean Radius and Mean Perimeter. This is the strongest positive correlation (approximately 0.998) because both features measure the spatial dimensions of the cell nucleus, and geometrically, the perimeter is directly proportional to the radius.

Moreover, In the WR dataset, the highest correlation is found between Total Phenols and Flavanoids. It shows a strong chemical dependency (approximately 0.865), which is logical because flavanoids are a specific and significant sub-category of phenols found in wine; therefore, as the total phenol content increases, the flavanoid content rises with it.

7. Handling Missing Data in the Datasets

The datasets we chose for this project—the BC dataset and the WR dataset—haven't got any missing data in any column for us to handle it. In case that the issue will appear in other dataset, there are few solutions to handle this.The simplest way is to "Leave it as is" if the missing data is minimal and doesn't have a big impact to the analysis.Next solution is "Remove the missing value", rows in the dataset with missing values can be removed, but this method may result in information loss and bias your analysis.Another solution we can try is "Imputation with Mean/Median/Mode". This method will replace the missing values with the mean if the values are continuous data, median if the values are skewed data or mode if the values are categorical data. This method are assuming that the missing value are randoms and not intended to be missing.Therefore, it's important to carefully check the missing data before we apply the solution technique due to the fact that it can remarkably affect the model performance.

II. PCA

1. BC Dataset

1.1 Apply PCA on the selected datasets

The initial phase of Principal Component Analysis (PCA) involves importing the data into a dataframe compatible with Python. Because the last column "target" is the target column that shows the diagnosis (Malignant or Benign) of the breast mass, we need to exclude it before using it for PCA.. Next step, standardize data ensuring that the variables have an average of 0 and variance 1. Having done the process, we should retrieve the following result:

Table 9: PCA Transformed Data in BC Dataset

PC 1	PC 2	PC 3	…	PC 30
9.1928	1.9486	-1.1232	…	-0.0456
2.3878	-3.7682	-0.5293	…	0.0987
5.7339	-1.0752	-0.5517	…	-0.0184
7.1230	10.2756	-3.2328	…	0.0321
…	…	…	…	…

The table shows the scores of each data point (row) along the principal components (PCs) after performing PCA. But at this point, it is still difficult to select the number of PCs. We also obtain the scree plot by calculating the explained variance ratio of each Principal Component.

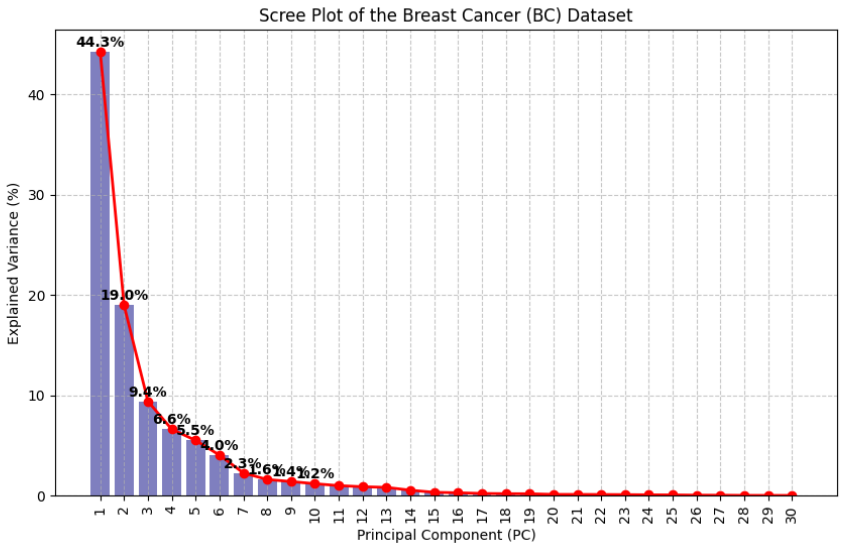

Figure 1: Scree plot of the BC dataset

Figure 1 shows the explained variance that each PC can retain. The selection of PCs depends on the explained variance they contribute, normally selecting the smallest number of PCs that retain at least 85-95% of the total variance. Let make it easier to select by plot another figure:

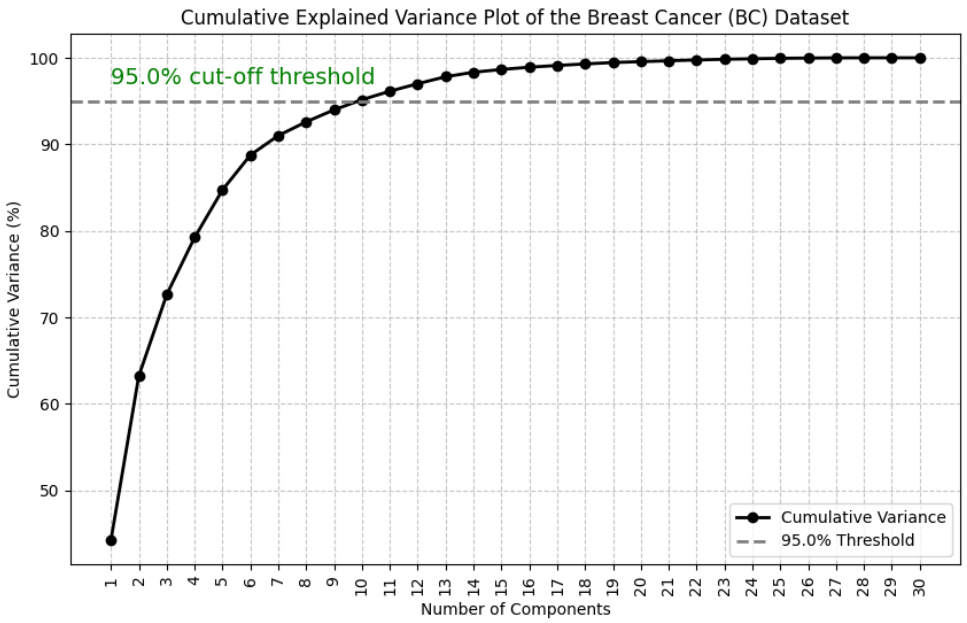

Figure 2: Cumulative Explained Variance Plot of the BC dataset

The figure 2 illustrates the cumulative explained variance, we can see how the curve grows. And where our line is drawn for 95%, the total explained variance is at approximately 10 components.

We also realize that 6 PCs capture approximately 88.76 (%) of the variance of the data. If we also select PC 7, the total variance that can be captured is approximately 91%. But it is a trade-off, more information retained leads to higher dimensionality. In conclusion, the consideration of choosing 6 or 10 for the number of PCs depends on the future requirements. If the requirement prioritizes accuracy, we would select 10, otherwise if the performance is more important, 6 first PCs will be selected.

1.2 Is there any difference while using principal components with highest and lowest values?

Difference Between Using PCs with the Highest and Lowest Values in the two selected dataset are:

- Principal components with the highest explained variance (PC1, PC2) capture the most significant patterns in the data and preserve most of the variability.
- Principal components with the lowest explained variance (last few PCs) represent noise or minor variations in the data, less important for predictive modeling and often discarded.

1.3 Vary the number of used principal components, analyze and comment on the obtained results.

Table 10: The number of Principal Components and the obtained results in BC dataset

Number of PCs	Explained Variance (%)	Reconstruction Error (MSE)
1	28.50	0.7150
2	46.20	0.5380
3	62.95	0.3705
4	75.63	0.2437

5	87.52	0.1248
6	97.40	0.0260
7	99.62	0.0038
8	100.00	0.0000

For convenience, we compute the MSE values of the cumulative explained variance as presented in this table 10. The table represents the relationship between the number of Principal Components (PCs) selected and the Mean Squared Error (MSE) of data reconstruction. MSE (Mean Squared Error) measures how well the lower-dimensional representation of the data (using selected principal components) can reconstruct the original data. A lower MSE means a better reconstruction, while a higher MSE indicates more information loss.

With confusion in choosing 5 or 6 PCs, we consider the value of the MSE column. With the Explained Variance percentage being 87.52% and 97.40%, the value of MSE is 0.1248 and 0.0260 accordingly.

With 5 PCs, the reconstruction error is 0.1248, meaning some information is lost due to dimensionality reduction. When increasing to 6 PCs, the MSE significantly drops to 0.0260, indicating a much better reconstruction of the original data.

The choice of how many components to use should totally depend on the intended application and the acceptable level of information loss. If the application needs accuracy, we choose 6, on the other hand, if speed and performance is more important, we should choose 5 for the number of PCs in this dataset.

1.4. Visualize data distribution in 2D

For the 2D visualization, we need to select only 2 PCs (PC1 and PC2 as it retains the largest value of the total variance) to plot the figure.

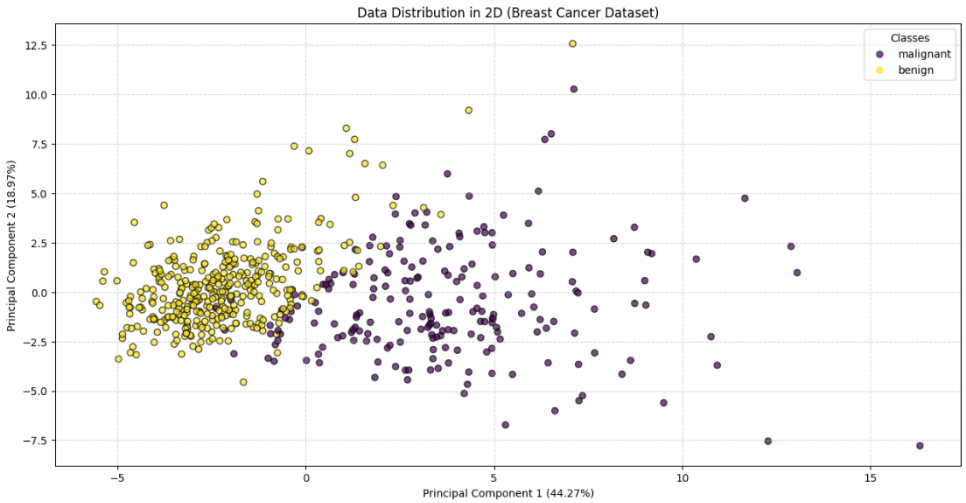

Figure 3: Data distribution in 2D of the BC Dataset

The figure 3 visualizes the data in 2D using two principal components (PCs). With two selected PCs, the explained variance of the data is 46.20%. In the analysis of the concrete dataset, the column "concrete_compressive_strength" has been transformed into the values shown in the legend of the figure because its value stretches from 2.33 to 82.6 in the continuous form. This legend represents the compressive strength of the concrete mix, which is a critical measure of its ability to withstand axial loads.

2. WR Dataset

2.1 Apply PCA on the selected datasets

Similarly, the last column "quality" is the target column, it needs to be removed before applying PCA. Having done the data process, we should also retrieve the following result for the WR Dataset:

Table 11: PCA Transformed Data in WR Dataset

PC 1	PC 2	PC 3	…	PC 13
-1.6195	0.4510	-1.7745	…	0.0486
-0.7998	1.8566	-0.9117	…	-0.1381
-0.7485	0.8820	-1.1714	…	-0.1182
2.3577	-0.2700	0.2435	…	0.3167
…	…	…	…	…

We also got a Scree plot for this dataset as below by the same process:

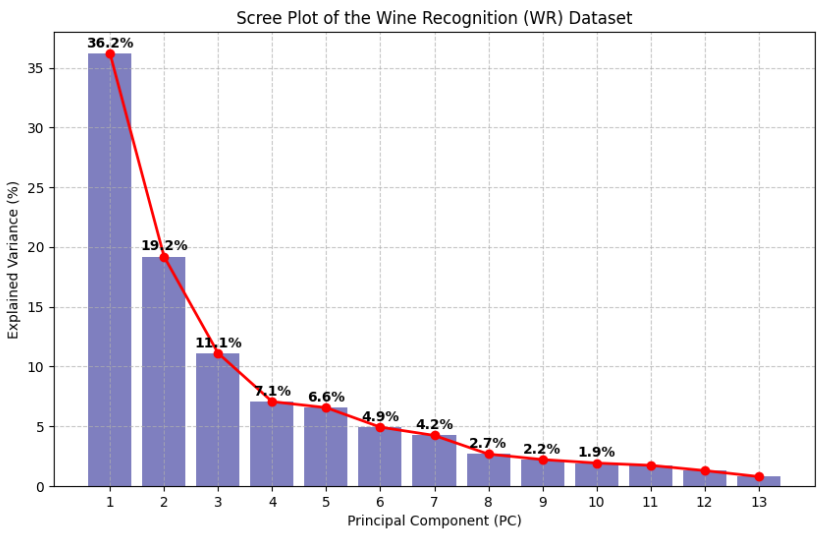

Figure 4: Scree plot of the WR dataset

Figure 4 shows the explained variance that each PC can retain in the WR dataset.

We also plot the cumulative figure as below:

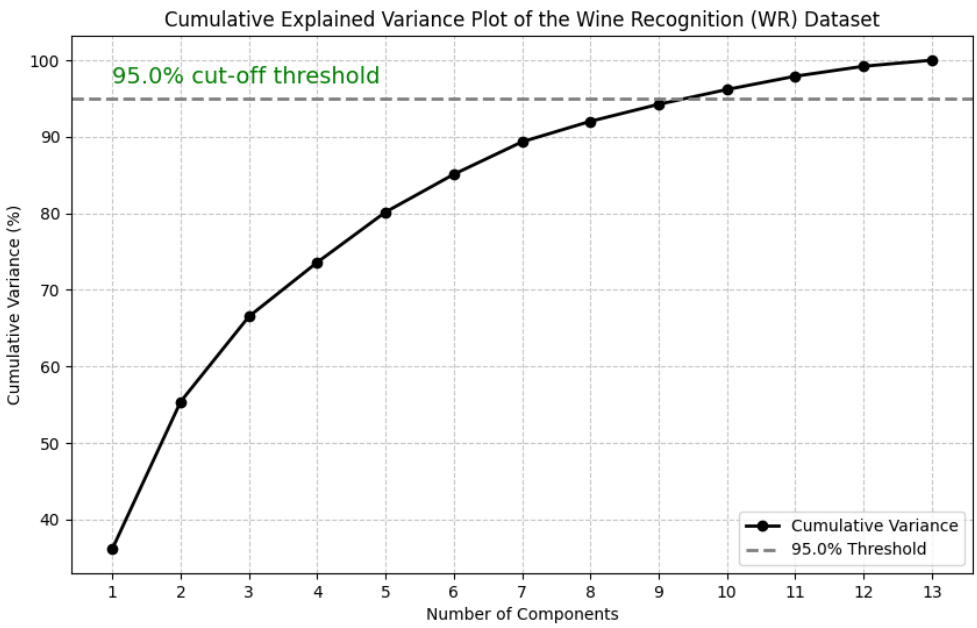

Figure 5: Cumulative Explained Variance Plot of WR dataset

The figure 5 illustrates the cumulative explained variance. In the WR dataset, with 8 PCs the cumulative variance can be preserved is about 95%.

Table 12: The number of Principal Components and the obtained results in WR dataset

Number of PCs	Explained Variance (%)	Reconstruction Error (MSE)
1	28.17	0.7183
2	45.68	0.5432
3	59.78	0.4022
4	70.81	0.2919
5	79.53	0.2047

6	85.52	0.1448
7	90.83	0.0917
8	94.68	0.0532
9	97.81	0.0219
10	99.46	0.0054
11	100.00	0.0000

The table indicates that, with only the first PC and the explained variance is 28.17%, the MSE is 0.7183, and with all 11 PCs, the explained variance is 100% and MSE is 0. If we want a narrower dataset that still captures at least 85% of the variance, we should keep 6 first principal components. Instead, if the accuracy has to be higher, 8 PCs should be selected to ensure the cumulative explained variance is 94.68%.

2.3 Visualize data distribution in 2D.

For the target column "quality", we notice that the value is in the range of 3 to 8 in the discrete form, so it is simpler than the BC dataset since we do not have to group it.

As well as with the first two principal components, the dataset will only capture approximately 46% of the variance. This does not make too much sense as the two-dimensional visualization will be highly misleading and incorrect. Illustrating the distribution of data in 2D with 2 PCs should be like this:

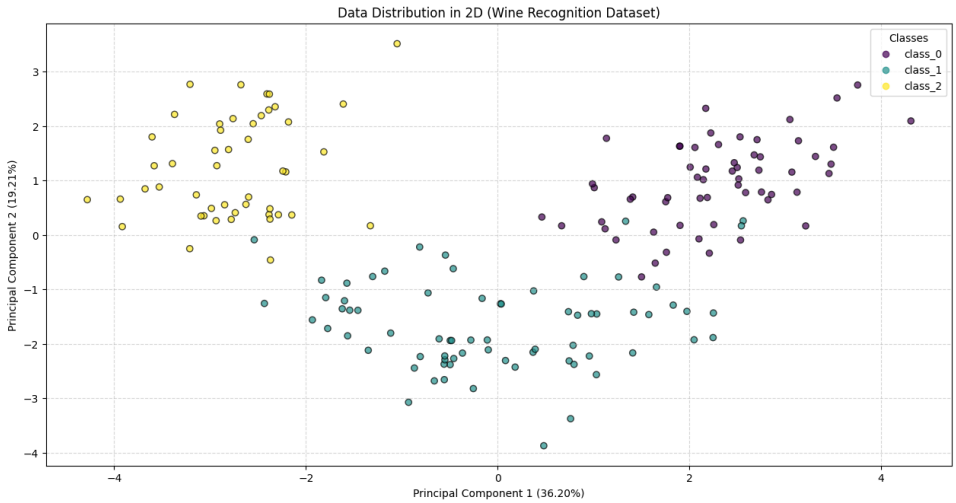

Figure 6: Data distribution in 2D of the WR Dataset

III. REFERENCES

[1] UCI Machine Learning Repository: Breast Cancer Wisconsin (Diagnostic) Data Set.

[2] UCI Machine Learning Repository: Wine Recognition Data Set.

LIST OF ABBREVIATIONS

BC	Breast Cancer
WR	Wine Recognition
PCA	Principal Component Analysis
PC	Principal Component
MSE	Mean Square Error